Abhinav Raj

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EDUCATION

University of Illinois Urbana-Champaign Champaign, IL
B.S. in Liberal Arts & Sciences, Statistics & Computer Science; Minor in Mathematics *Expected May 2029*

- GPA: 4.0/4.0; Dean's List.
- Coursework: CS 225 Data Structures (In Progress), CS 233 Computer Architecture (In Progress), CS 173 Discrete Structures, Linear Algebra, Probability and Statistics.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, MySQL.
- **AI/ML Tools:** Ollama, Anthropic API.
- **Developer Tools:** Git, Android Studio.

EXPERIENCE

ACM@UIUC — SIGNLL Champaign, IL
Member *Aug 2025 – Present*

- Measured a statistically robust 22.4-point accuracy gap between Claude Haiku 4.5 at 80.0% and the best local 3B model at 57.6% (95% CI [16.8, 28.0]) on FinQA financial numerical reasoning, with Sonnet 4.6 highest at 85.2%.
- Benchmarked Llama 3.2, Qwen 2.5, and Phi 3.5 locally through Ollama at 8-bit quantization on a single consumer GPU, at zero marginal API cost.
- Found the gap collapses on FOMC monetary-policy stance classification — a 17.7-point macro-F1 spread versus 3.5 on numerical reasoning — showing the deficit is specific to multi-step arithmetic reasoning rather than general financial language understanding.
- Built a unit-aware answer-extraction and dual-gold scoring layer reaching 99.6% parse coverage, and audited scoring tolerance across 434 boundary cases to confirm small-model errors were reasoning failures rather than parsing artifacts.
- Measured quality, latency, and cost using multi-tolerance accuracy, bootstrap confidence intervals, and paired-bootstrap significance tests; packaged the pipeline as a config-driven harness with persisted per-item outputs for reproducible re-scoring.

Financial Engineering Club Champaign, IL
Member *Feb 2026 – Present*

- Built a Python research pipeline comparing Buy and Hold, 252-day Time-Series Momentum, Directional Change, and Multi-Threshold Directional Change strategies on SPY, QQQ, TLT, and GLD daily ETF data from 2010 to 2026; implemented one-day-lagged backtests with normalized exposures and 0, 5, and 10 bps transaction-cost checks.
- Found no active rule beat passive exposure — Buy & Hold returned 649.70% (1.01 Sharpe) against momentum's 76.62% (0.41) — and that both directional-change variants lost money before any transaction costs (−48.37% and −53.28%), isolating signal quality rather than trading friction as the cause of failure.
- Traced the failure to trading intensity: the event-based rules generated 1,540 and 1,849 position changes against momentum's 308, at five to six times the turnover, producing maximum drawdowns of −69.27% and −74.50%.
- Showed the multi-threshold extension degraded rather than stabilized the signal, underperforming the single-threshold rule it was designed to make robust across every tested cost level.

Solar Car Team Illinois Champaign, IL
Strategy Team Member *Aug 2025 – Dec 2025*

- Identified anomalous battery discharge patterns across 240K+ telemetry records using Python and SQL, informing race strategy adjustments and data-quality pipeline improvements.

PROJECTS

SilentSync 2023 - 2025

- Built a native Android app for workplace communication accessibility for deaf employees, supporting speech-to-text conversation capture, quick-phrase shortcuts, and an ASL fingerspelling practice mode with live camera feedback.
- Ran all machine learning on-device with no backend, so the app functions fully offline and no user data leaves the phone; distributed directly to users and local business partners and iterated on their feedback.